Preconditions

`this->valid()` would return `true`.

Effects

If the asynchronous result associated with `*this` contains a deferred function arising from a call to `std::async` that has not yet started execution, returns immediately without blocking. Otherwise, blocks until the asynchronous result associated with `*this` is *ready* or the time period specified by `relative_time` has elapsed.

Returns

`std::future_status::deferred` if the asynchronous result associated with `*this` contains a deferred function arising from a call to `std::async` that hasn't yet started execution, `std::future_status::ready` if the asynchronous result associated with `*this` is *ready*, `std::future_status::timeout` if the time period specified by `relative_time` has elapsed.

Note

The thread may be blocked for longer than the specified duration. Where possible, the elapsed time is determined by a steady clock.

Throws

Nothing.

Std::Shared_Future::Wait_Until Member Function